Time-Aware Network Centrality Measures & Link Prediction
(UN-DIRECTED NETWORK)

① **Dataset**: Stack Overflow Temporal Network

Nodes: 2601977

Temporal Edges: 63497050

Time Span: 2774 days

② **Temporal Edges**:

$$e_{ij}(t) = \langle \underbrace{v_i}_{\text{source node}}, \underbrace{v_j}_{\text{target node}}, \underbrace{t}_{\text{timestamp}} \rangle$$

source_id target_id timestamp

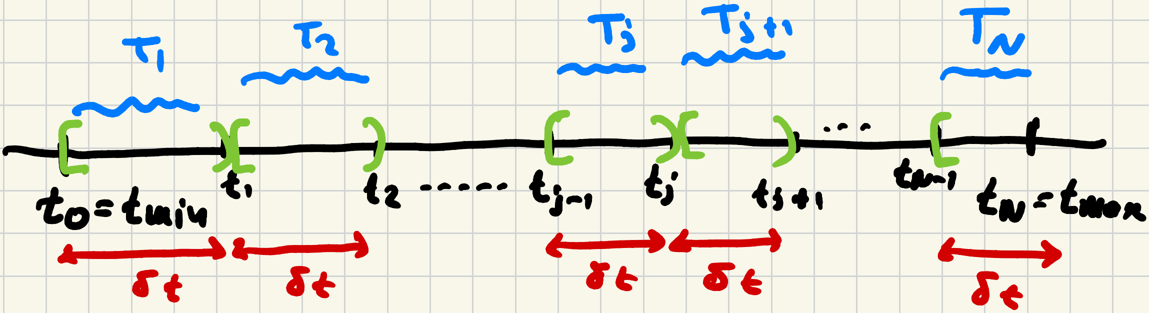
N

③ Complete Time Period:

$$T = [t_{min}, t_{max}]$$

$$\boxed{\Delta T = t_{max} - t_{min}}$$

④ Time Instances & Time Periods



N : # OF TIME PERIODS

$$\boxed{\delta t = \frac{\Delta T}{N} = \frac{t_{max} - t_{min}}{N}}$$

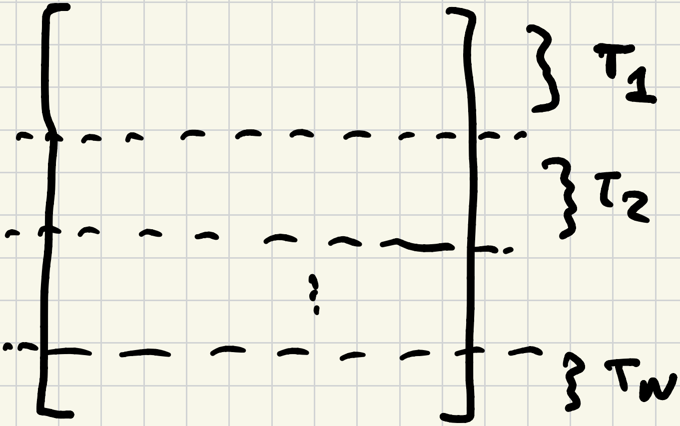
$$t_j = t_{min} + j \cdot \delta t, \quad 0 \leq j \leq N$$

For $j = 0 \Rightarrow t_0 = t_{min}$

For $j = N \Rightarrow t_N = t_{min} + \cancel{N} \cdot \frac{t_{max} - t_{min}}{\cancel{N}} \Rightarrow$

$$t_N = t_{min} + t_{max} - t_{min} \Rightarrow$$

$$t_N = t_{max}$$



$$\boxed{T_j = \begin{cases} [t_{j-1}, t_j) \,, & 1 \le j \le N-1 \\ [t_{j-1}, t_j] \,, & j = N. \end{cases}}$$

⑤ <u>Temporal Graphs</u> : $\forall j : 1 \le j \le N$

$$\boxed{G[t_{j-1}, t_j] = \left(V[t_{j-1}, t_j], E[t_{j-1}, t_j]\right)}$$

Network / Graph for time period T_j.

$$\boxed{E[t_{j-1}, t_j] = \{ e_{ij}(t) : t_{j-1} \le t \le t_j \}}$$

$V[t_{j-1}, t_j]$: The set of vertices which appear as edge-points for edges in $E[t_{j-1}, t_j]$

$$G_1 \longrightarrow G_2 \Rightarrow G_3 \Rightarrow \dots \Rightarrow G_N$$

$$G_k = (V_k, E_k)$$

$G_k \rightarrow$ Visualization

$G_k \rightarrow$ $\boxed{\text{Degree Centrality Distribution}}$

$G_k \rightarrow$ Katz Centrality Distribution

$G_k \rightarrow$ Betweenness Centrality Distribution

⋮

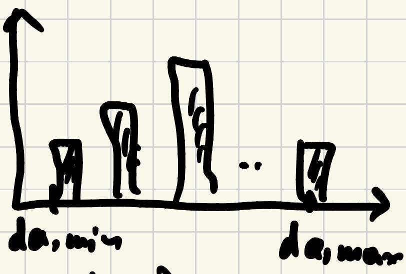
$G_k \rightarrow$ Centrality Measure Distribution

$$G_1 : \underline{P_1} = [P_1(d_{1,min}), \dots, P_1(d_{1,max})]$$

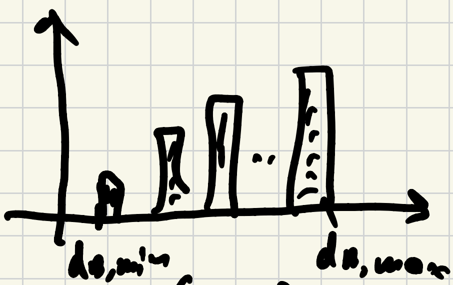
⋮

$$G_k : \underline{P_k} = [P_k(d_{k,min}), \dots, P_k(d_{k,max})]$$

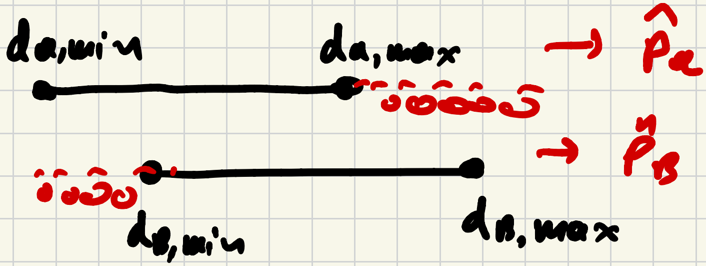
⋮



(G_a)



(G_b)



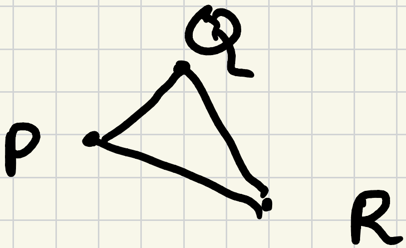
$$d_{min} = \min(d_{a,min}; d_{B,min})$$

$$d_{max} = \max(d_{a,max}; d_{B,max})$$

$$D_{KL}(\hat{P}_a; \hat{P}_B) = \sum_{d=d_{min}}^{d=d_{max}} \hat{P}_a(d) \cdot \log \frac{\hat{P}_a(d)}{\hat{P}_B(d)}$$

$$\log(x) \equiv \log(x+\varepsilon) \qquad \varepsilon = 10^{-4}$$

$$\frac{a}{x} \equiv \frac{a}{x+\varepsilon}, \qquad \varepsilon = 10^{-4}$$



$$d_{PQ} + d_{QR} \leq d_{PR}$$

⑥ Network Evolution from $T_j \longrightarrow T_{j+1}$

$$G[t_{j-1}, t_j] \Longrightarrow G[t_j, t_{j+1}]$$

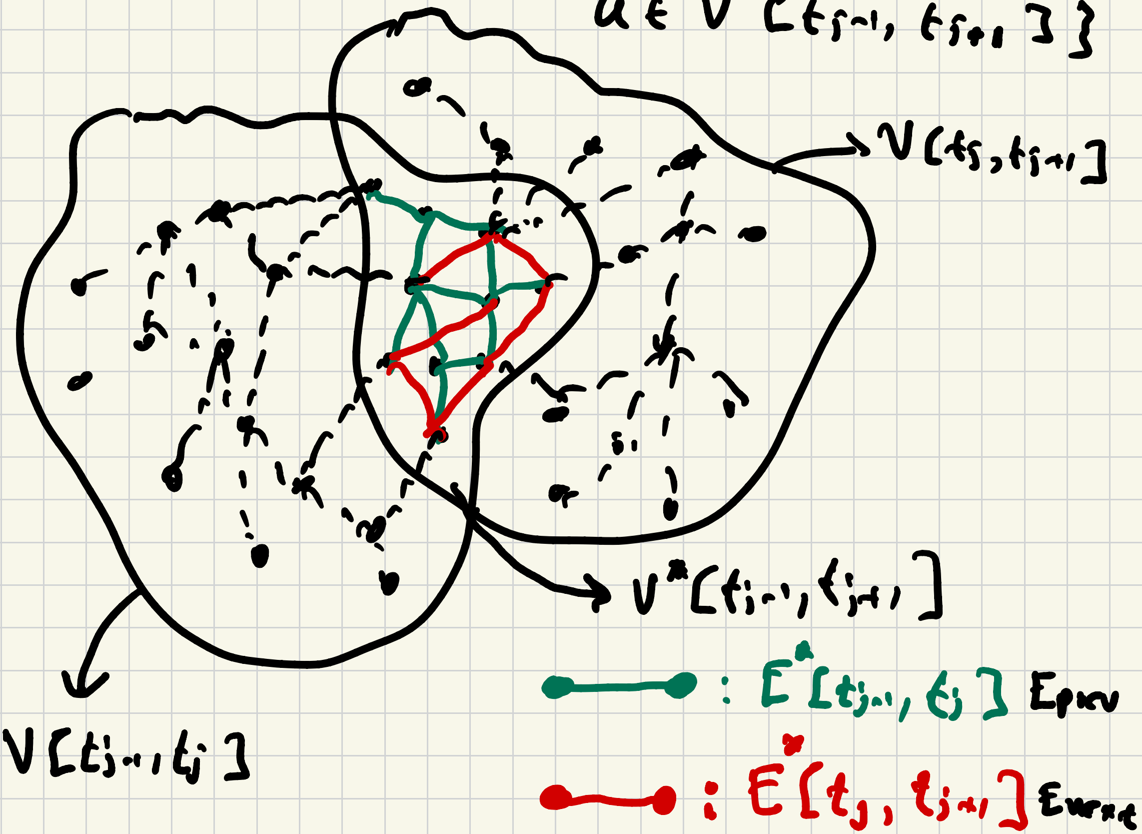
$$V[t_{j-1}, t_j] \Longrightarrow V[t_j, t_{j+1}]$$

$$E[t_{j-1}, t_j] \Longrightarrow E[t_j, t_{j+1}]$$

$$\boxed{V^*[t_{j-1}, t_{j+1}] = V[t_{j-1}, t_j] \cap V[t_j, t_{j+1}]}$$

$$E^*[t_{j-1}, t_j] = \{(v,u) \in E[t_{j-1}, t_j] : v \in V^*[t_{j-1}, t_{j+1}] \wedge u \in V^*[t_{j-1}, t_{j+1}]\}$$

$$E^*[t_j, t_{j+1}] = \{(v,u) \in E[t_j, t_{j+1}] : v \in V^*[t_{j-1}, t_{j+1}] \wedge u \in V^*[t_{j-1}, t_{j+1}]\}$$



Similarity Measures For Pairs of Nodes

Training : $G_{train} = (V^*[t_{j-1}, t_{j-1}], E^*[t_{j-1}, t_j])$

$\forall (u,v) \in V^* \times V^*$: $\underline{\varphi}_{uv} = [S^{SP}_{uv}, S^{CN}_{uv}, S^{JC}_{uv}, S^{AA}_{uv}, S^{PA}_{uv}]$

I) $S_{sp}(u,v) = \dfrac{1}{d_{sp}(u,v)}$

SP: SHORTEST PATH
CN: COMMON NEIGHBOURS
JC: JACCARD COEF
AA: ADAMIC/ADAR
PA: Preferential Attachment

(i): IF $d_{sp}(u,v) = 1 \Rightarrow S_{sp}(u,v) = 1$

(ii): IF $d_{sp}(u,v) = \infty \Rightarrow S_{sp}(u,v) = 0$

II) $S_{CN}(u,v) = |\Gamma(u) \cap \Gamma(v)| \in [0, \max(|\Gamma(u)|, |\Gamma(v)|)]$

where $\Gamma(z)$: the neighborhood of node z.

III) $S_{JC}(u,v) = \dfrac{|\Gamma(u) \cap \Gamma(v)|}{|\Gamma(u) \cup \Gamma(v)|} \in [0,1]$

IV) $S_{AA}(u,v) = \displaystyle\sum_{z \in \Gamma(u) \cap \Gamma(v)} \frac{1}{\log_2 |\Gamma(z)|}$

(i): $\Gamma(u) \cap \Gamma(v) = \emptyset$
$S_{AA} = 0$

(ii): $|\Gamma(z)| = 1$?

$|\Gamma(z)| \geq 2$

$u \in \Gamma(z)$
$v \in \Gamma(z)$

v) $S_{PA}(u,v) = |\Gamma(u)| \cdot |\Gamma(v)|$

$$E_0 \equiv V^* \times V^* \quad (\text{ΟΛΕΣ ΟΙ ΠΙΘΑΝΕΣ ΑΚΜΕΣ})$$

$$E_0 = \underbrace{E_{prev} \cup E_{next}}_{E_{ex}} \cup E_{nonex}$$

$$E_{ex} \cup E_{nonex}$$

$$|E_{ex}| << |E_{nonex}|$$

$$\begin{tikzcd}
 & E_{ex} \arrow[dl] \arrow[dr] & & & E_{nonex} \arrow[dl] \arrow[dr] & \\
E^{ex}_{train} \arrow[d, phantom, "\equiv" rotate=90] & & E^{ex}_{test} \arrow[d, phantom, "\equiv" rotate=90] & E^{train}_{nonex} & & E^{test}_{nonex} \\
E_{prev} & & E_{next} & & &
\end{tikzcd}$$

$$E_{train} = E^{ex}_{train} \cup E^{train}_{nonex}$$

$$E_{test} = E^{ex}_{test} \cup E^{test}_{nonex}$$

Balancing Conditions:

$$|E^{train}_{nonex}| \approx |E_{train}| = |E_{prev}|$$

$$|E^{test}_{nonex}| \approx |E_{test}| = |E_{next}|$$

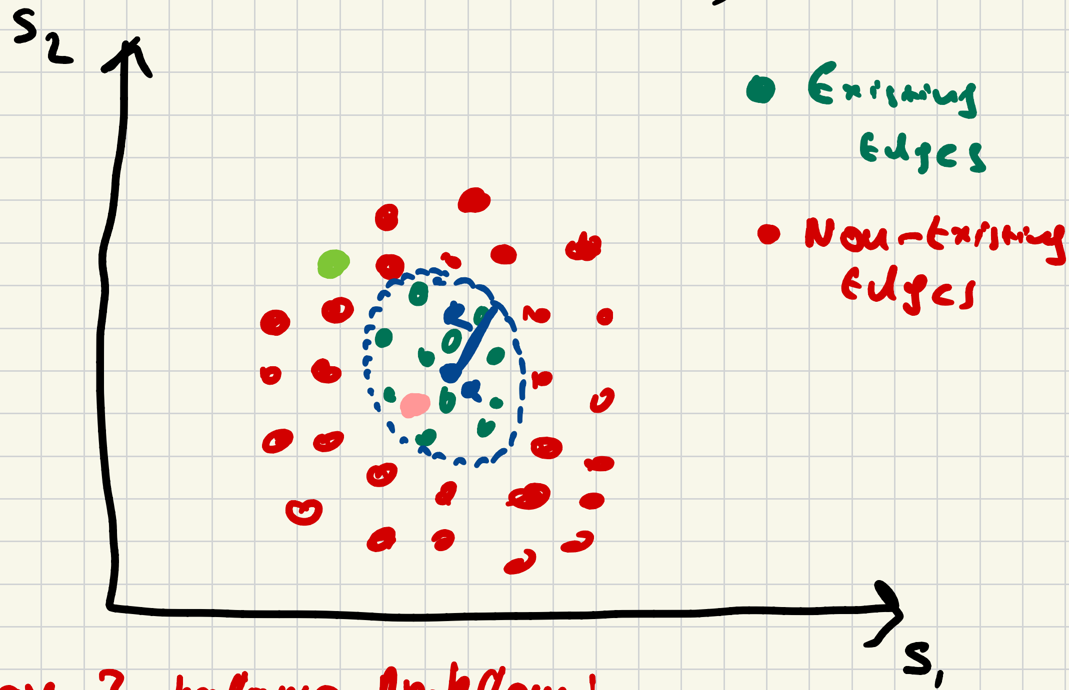
C_1 vs C_2

$$P(C_1) >> P(C_2)$$

$P(C_1)$ → Majority Class

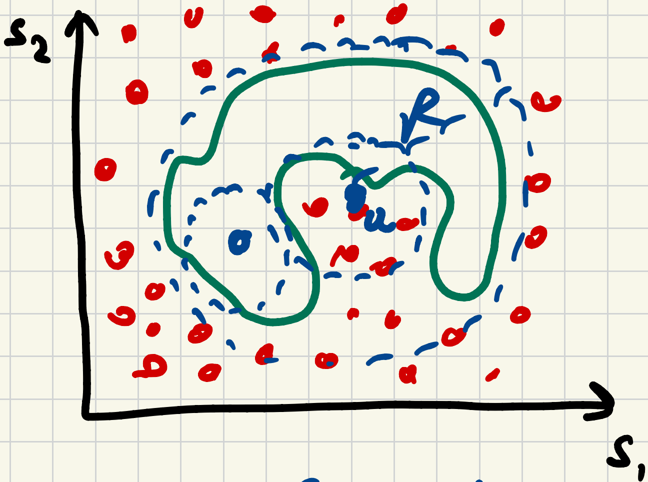
$P(C_2)$ → Minority Class

$$\tilde{P}(C_1) \approx \tilde{P}(C_2)$$

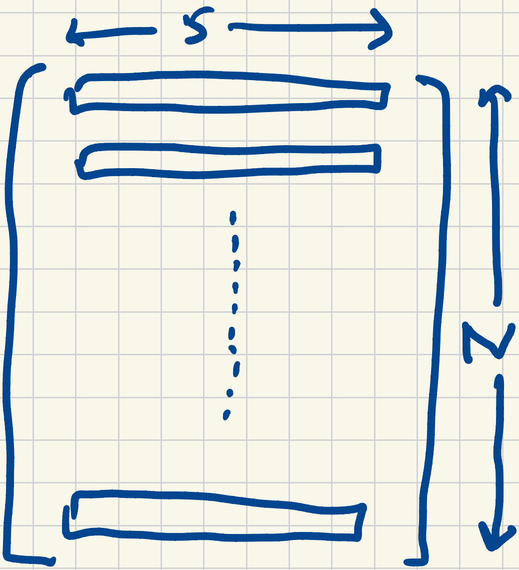


Class Imbalance Problem!

One Class Learning (One Class SVMs)



$\underline{\underline{X}} =$



$$\underline{y} = \begin{bmatrix} +1 \\ +1 \\ \vdots \\ +1 \\ \vdots \\ -1 \\ -1 \\ \vdots \\ -1 \end{bmatrix}$$

$N = |E_{train}|$

$+1$: existing edge

-1: nonexisting edge

$$F_{NN} : \mathbb{R}^{s} \longrightarrow \{-1, +1\}$$